Deep Learning on Embedded Systems (esp. NVIDIA Jetson)

Application Area: Drones

Sparsh Mittal

Drone Racing

Jung et al., “Perception, Guidance, and Navigation for Indoor Autonomous Drone Racing Using Deep Learning”, LRA, 2018

Drone racing competition

- Drone racing competition held during IROS 2016 conference requires a drone to fly through a series of 26 gates with 1.3m x 1.3m square as fast as possible.
- **Challenge:** techniques based on color and gate-size do not work well since multiple gates situated behind each other appear to overlap.

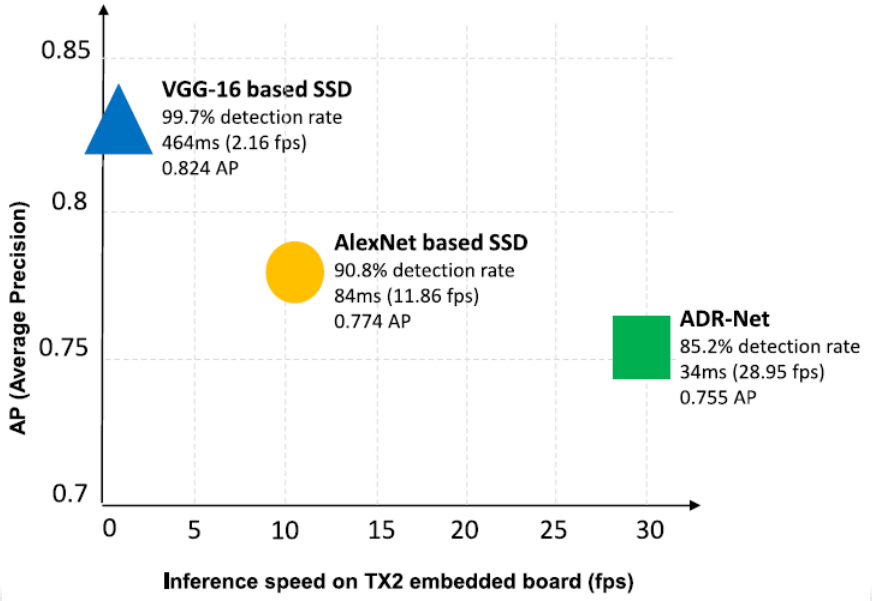
Use of sensors

- Drone uses a LIDAR for measuring altitude, an optical flow camera for measuring ground speed and a stereo camera for measuring odometry.
- Odometry and velocity data are fed to the ``unscented Kalman filter'' which fuses above information with IMU sensor to obtain more accurate value of velocity and position.
- All processing, including execution of unscented Kalman filter, happens on TX2.

- Gate detection using SSD (Single Shot MultiBox Detector) algorithm

Optimizations to SSD algorithm

- SSD has high latency => replace its base network from VGG-16 to AlexNet and remove one FC layer
- Only one gate needs to be detected => some non-essential high-level feature layers are removed

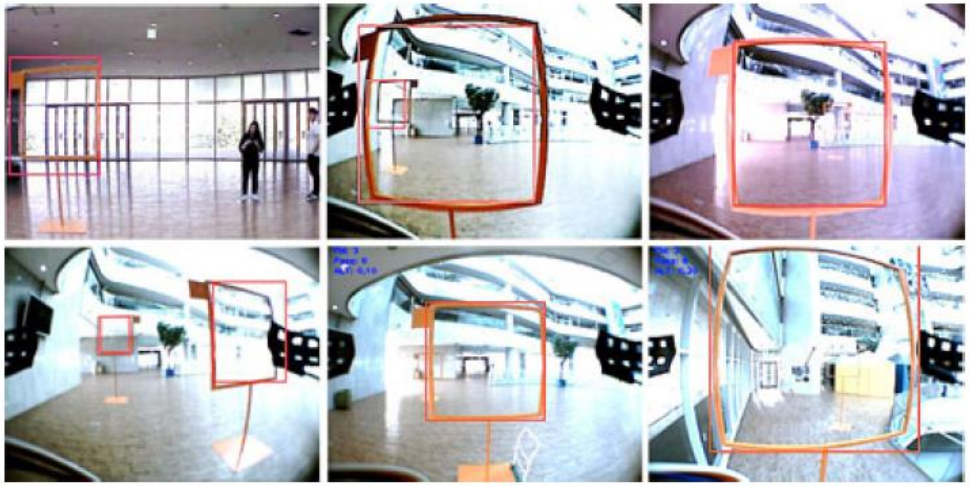


Recovering precision loss

- Box sizes used in SSD are ascertained based on their dataset, since original SSD is trained on PASCAL VOC
- e.g., since gate-size is not small, box size was increased
- On detecting multiple gates, the closest gate is selected which has highest probability

- On detecting a gate, their technique extracts its center point and runs the "line-of-sight guidance" algorithm for making drone follow center point.
- The algorithm seeks to minimize the lateral error between the drone and gate center.

Results



- Their drone passes through all the gates with near-zero lateral error
- can also pass through a moving gate
- Limitation: next gate to be traversed has to be in view at all times => their technique is useful only in simple environments